Indian Institute of Science Education and Research Kolkata
Department of Mathematics and Statistics
Subject: Probability and Statistics-I (MA2104)
Class Test-1 Full marks: 10
Date (Time) of Test : 21.09.2025 (11:00AM) Max. time: 45 minutes

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Instructions: There are three questions in all. Answer all the questions. In order to get full credit, all the intermediate steps must be provided and the answers must be given in its most simplified forms. Every answer must be explained with proper reasoning. The notations have their usual meanings. The number inside the square brackets at the right of each question denotes the marks corresponding to that question.

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1. In a multiple choice question, there are four options of which only one answer is correct. If a student answers it correctly s/he gets 1 mark, not answering brings 0 marks whereas wrong answer fetches $(-1/3)$ for him/her. If there are 10 such questions, find the expected marks for the student. [3]

2. A biased coin with probability of head 0.6 and a fair die are tossed together. Let X denote the number coming up on the die whereas Y takes the value 1 if head comes and takes value 2 if tail appears. Find $Cov(X, Y)$. [1]

3. Find the parameters of a binomial distribution if its expectation is 10 and variance is 7. [2]

4. Show that
$$F_X(x) + F_Y(y) - 1 \leqslant F_{X,Y}(x, y) \leqslant \sqrt{F_X(x)F_Y(y)}$$
for any x and y, where F_Z is the cdf of the random variable Z. [4]

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Indian Institute of Science Education and Research Kolkata

Department of Mathematics and Statistics

Subject: Probability and Statistics-I (MA2104)

Class Test-2 Full marks: 10

Date (Time) of Test : 29.10.2025 (9:00AM)

Max. time: 45 minutes

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Instructions: There are five questions in all. Answer all the questions. In order to get full credit, all the intermediate steps must be provided and the answers must be given in its most simplified forms. Every answer must be explained with proper reasoning. The notations have their usual meanings. The number inside the square brackets at the right of each question denotes the marks corresponding to that question.

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1. Find the A.M. of $5, 55, 555, \ldots$ up to n terms. [2]

2. If the mean square deviation of a variable x about 3 is 20 and the mean of x is 5, calculate the minimum value of root mean square deviation of x. [2]

3. If the relationship between two variables x and y is $2x+3y = 7$ and the first quartile of x is 9, then find the third quartile of y. [2]

4. Let the A.M. and the C.V. of a random variable X are 10 and 20% respectively. Find the variance of $(4 - 3X)$. [2]

5. Let a variable X take only two values x_1 and x_2 with respective frequencies f_1 and f_2. If [2]

$$V(X) = \frac{(x_1 - x_2)^2 f_1 f_2}{49},$$

find the total frequency $(f_1 + f_2)$.

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